

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

3-amaliy mashg'ulot. Filtrlash, shovqin va diffuziya

2026–2027 o'quv yili

Muhit (setup)

Python 3.11 yoki 3.12, faqat CPU. Versiyalar qat'iy belgilangan (`requirements.txt`): `numpy==2.4.4`, `scipy==1.17.1`, `pandas==3.0.2`, `scikit-learn==1.8.0`, `scikit-image==0.26.0`, `matplotlib==3.10.9`, `pillow==12.2.0`, `opencv-python-headless==4.13.0.92`, `torch==2.14.0`, `torchvision==0.29.0`.
O'rnatish: `python -m venv .venv`, so'ng `pip install -r requirements.txt`. Import nomi: `import cv2`. Tasodifiylik: `np.random.default_rng(42)`. Barcha rasmlar `skimage.data` ichida (internet shart emas); natija rasmlari `./out/` ga saqlanadi.

Mavzu. Chiziqli filtrlar: konvolyutsiya va kross-korrelyatsiya, Gaussian filtr, ajratiluvchanlik (separability), integral tasvir, chegara shartlari (border modes); shovqin modellari (Gauss, tuz-qalampir); chiziqli filtrlar: median, bilateral, NLM; PSNR/SSIM bilan σ tanlash; issiqlik tenglamasi va barqarorlik; Perona–Malik anizotrop diffuziyasi; unsharp masking; Wiener filtri; CNN qatlami. **Maqsadlar.** Konvolyutsiyani qo'lda va dasturda hisoblash; shovqin turiga mos filtr tanlash va natijani o'lchash; Gauss silliqlash va diffuziya ekvivalentligini hamda oshkor sxema barqarorligini sonli tekshirish; Perona–Malik parametrlarini sozlash. **Baholash.** 1–16-topshiriqlar (qo'lda hisoblash va nazariya) — har biri 2 ball (32); 17–30-topshiriqlar (kod) — har biri 4 ball (56); kod sifati va hisobot — 12 ball. Jami 100 ball. Kod `python solutions.py` bilan xatosiz, takrorlanuvchan ishlashi shart.

Belgilashlar. Korrelyatsiya $(I \star K)(u, v) = \sum_{i,j} K(i, j)I(u + i, v + j)$, konvolyutsiya $(I * K)(u, v) = \sum_{i,j} K(i, j)I(u - i, v - j)$; $S_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$, $L = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$; Perona–Malik: $c_1(g) = e^{-(g/k)^2}$, $c_2(g) = 1/(1 + (g/k)^2)$.

1. Qo'lda hisoblash, keltirib chiqarish va nazariy savollar

- Konvolyutsiya qo'lda.** $P = \begin{pmatrix} 12 & 15 & 20 \\ 14 & 18 & 25 \\ 16 & 22 & 30 \end{pmatrix}$ yamog'i markazida: (a) 3×3 o'rtachalash; (b) $\frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$; (c) S_x korrelyatsiya va konvolyutsiya bilan. Nega (c) da ishora almashadi?
- Laplasiyan.** $I = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 3 & 1 & 2 & 2 \\ 0 & 2 & 4 & 1 \\ 1 & 0 & 1 & 3 \end{pmatrix}$ ga L ni qo'llang (faqat to'liq oyna sig'adigan “valid” 2×2 natija).
- Gauss yadrosi.** $\sigma = 0,8$, $x \in \{-2, \dots, 2\}$: normallangan 1D yadroni toping (4 xona). $\sigma = 2,5$ da $[3\sigma]$ qoidasi bo'yicha yadro o'lchami, 2D va ajratilgan holda piksel boshiga ko'paytirishlar soni.
- Ajratiluvchanlik.** Quyidagilarning rangini toping va mumkin bo'lsa ab^T ko'rinishda yozing: $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$, $\begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. Rang 1 $\iff$ ajratiluvchi ekanini isbotlang.
- Gausslar kompozitsiyasi.** $G_{1,2}$ va $G_{1,6}$ ketma-ket qo'llansa, ekvivalent σ ? Oshkor issiqlik sxemasida $\Delta t = 0,25$ bilan $\sigma = 3$ ga necha qadamda erishiladi ($\sigma^2 = 2N\Delta t$)?
- Chegara shartlari.** $x = (1, 2, 3, 4, 5)$ ga $(1, 2, 1)/4$ yadro: REFLECT_101, REPLICATE, CONSTANT (0), WRAP rejimlarida natijani toping. Qaysi biri chiziqli signalni chetda ham saqlaydi?
- Median.** 3×3 oyna $(12, 14, 13, 0, 255, 15, 13, 14, 12)$: o'rtacha va median. Tuz-qalampir zichligi $\rho = 0,2$ bo'lsa, oynada kamida 5 ta buzilgan piksel bo'lish ehtimoli ($X \sim \text{Bin}(9, \rho)$)?
- Bilateral og'irliklar.** 1D signal $(20, 20, 20, 100, 100)$, markaziy piksel, $\sigma_s = 1$, $\sigma_r = 25$. Fazoviy va qiymat og'irliklarini, bilateral natijani hamda oddiy Gauss (faqat fazoviy) natijani toping.
- Chastota javobi.** 1D quti $(1, 1, 1)/3$ uchun $H(\omega) = (1 + 2\cos\omega)/3$ ni keltirib chiqaring; nol va manfiy sohani, $H(\pi)$ ni toping. Uzluksiz Gauss ($\sigma = 1$) uchun $e^{-\sigma^2\omega^2/2}$ ni $\omega = \pi$ da hisoblang.
- Integral tasvir.** $I = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{pmatrix}$ uchun S ni tuzing va 1–2-qatorlar, 1–3-ustunlar (0 dan sanaganda) yig'indisini 4 ta murojaat bilan toping.
- Issiqlik tenglamasi qadami.** $\begin{pmatrix} 30 & 30 & 30 \\ 30 & 70 & 30 \\ 30 & 30 & 30 \end{pmatrix}$ markazida $L * u$ va $\Delta t = 0,15$ dagi u^1 ni toping. Chiziqli naqsh (masalan, $P_{ij} = 10i + 20j$) uchun $L * P$ nimaga teng?
- Barqarorlik.** Fon Neyman usuli bilan 2D oshkor sxema uchun $\lambda = 1 - 4\Delta t(\sin^2 \frac{a}{2} + \sin^2 \frac{b}{2})$ ni keltirib chiqaring. $\Delta t = 0,35$ da eng tez o'suvchi moda ko'paytuvchisi va u 10^6 ga necha qadamda yetadi?
- Perona–Malik.** c_1 va c_2 ni $g = k/2$, $k, 2k$ da hisoblang. 1D $(10, 10, 50, 50)$, $k = 20$, $\Delta t = 0,25$, c_1 : belgilangan pikselning bir qadamdan keyingi qiymati; issiqlik tenglamasida-chi? $\Phi(g) = g c_1(g)$ ning maksimum nuqtasini toping.

14. **Unsharp masking.** Pogʻona $(0, 0, 100, 100)$ ni $(1, 2, 1)/4$ bilan silliqlab (chetda `edge`), $\lambda = 1,5$ da $I_s = I + \lambda(I - G * I)$ ni toping. Qanday xavf paydo boʻladi?
15. **CNN qatlami.** Kirish $128 \times 128 \times 3$, 32 ta 5×5 yadro, $P = 2$, $S = 2$: chiqish oʻlchami va parametrlar soni. `nn.Conv2d` konvolyutsiyami yoki korrelyatsiya? CNN uchun bu nega muhim emas?
16. **Wiener va oʻrtachalash.** $H = S_I / (S_I + \sigma_n^2)$: $\sigma_n^2 = 100$, $S_I = 400$ va $S_I = 25$ da H . $\sigma_n = 20$ shovqinli tasvirda 5×5 oʻrtachalash shovqin standart chetlanishini qanchaga kamaytiradi va nega?

2. Dasturlash topshiriqlari

Barcha kodlar bitta `solutions.py` faylida: har bir topshiriq — alohida funksiya (`task17()`, ..., `task30()`) va `main()`. Umumiy qism:

```
import numpy as np, cv2, torch, torch.nn.functional as F, skimage.data
from scipy import ndimage
from skimage.metrics import peak_signal_noise_ratio as psnr, structural_similarity as ssim
def noisy_images(seed=42):
    # 22, 23-topshiriqlar
    rng = np.random.default_rng(seed); g = skimage.data.camera()
    gn = np.clip(np.round(g + rng.normal(0, 20, g.shape)), 0, 255).astype(np.uint8)
    sp = g.copy(); m = rng.random(g.shape); sp[m < 0.05] = 0; sp[m > 0.95] = 255
    return g, gn, sp
    # PSNR/SSIM: data_range=255
```

Hujjatlar: https://docs.opencv.org/4.x/d4/d86/group_imgproc_filter.html, https://docs.opencv.org/4.x/d4/d13/tutorial_py_filtering.html, https://docs.opencv.org/4.x/d5/d69/tutorial_py_non_local_means.html, <https://scikit-image.org/docs/stable/api/skimage.metrics.html>, <https://docs.scipy.org/doc/scipy/reference/ndimage.html>, <https://pytorch.org/docs/stable/generated/torch.nn.Conv2d.html>.

17. **Korrelyatsiya va konvolyutsiya.** `correlate2d(img, K)` (chegara `REFLECT_101`, yaʼni `np.pad(..., mode='reflect')`), yadro siljishlari boʻyicha sikl) va `convolve2d` (agʻdarilgan yadro). `camera()` (`float64`), S_x : `cv2.filter2D` va `ndimage.convolve(..., mode='mirror')` bilan farq; $\max|\text{corr} + \text{conv}|$, $\max|\text{corr} - \text{conv}|$; 1-topshiriq yamogʻida tekshiring.
18. **Gauss yadrosi va ajratish.** `gaussian_kernel1d(sigma)` (radius $\lceil 3\sigma \rceil$, normallangan) va `cv2.getGaussianKernel` farqi, $\sigma \in \{0,8; 1,5; 5\}$; amallar soni. $\sigma = 5$ da `filter2D` (2D yadro) va `sepFilter2D` farqi, rang. $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$: SVD, singulyar qiymatlar va eng yaxshi rang-1 yaqinlashish xatosi (Frobenius).
19. **Kompozitsiya.** `GaussianBlur` ($\sigma = 3$, keyin $\sigma = 4$) va bitta $\sigma = 5$ (`borderType=cv2.BORDER_REFLECT`); chetdan 30 px ichkarida oʻrtacha va maksimal farq.
20. **Chegara rejimlari.** `blur_border(img, sigma, border)`: `cv2.copyMakeBorder` (radius $\lceil 4\sigma \rceil$) + `sepFilter2D` ($\sigma = 3$) + kesish (`GaussianBlur` `WRAP` ni qoʻllamaydi). `REFLECT`, `REPLICATE`, `WRAP`, `CONSTANT` ni `REFLECT_101` bilan solishtiring: 0-qatorda oʻrtacha, maksimal va chetdan ≥ 12 px ichkaridagi farq. 6-topshiriqni kodda tekshiring.
21. **Integral tasvir.** `box_filter_integral(img, r)` — `cv2.integral` va 4 murojaat bilan $(2r+1)^2$ quti oʻrtachasi (ichki soha). $r \in \{1, 7, 25\}$ da `cv2.blur` bilan farq; 10-topshiriqni tekshiring.
22. **Shovqin va filtr tanlash.** `noisy_images()`: `blur` 3×3 , `GaussianBlur` $\sigma = 1$, `medianBlur` 3, `bilateralFilter(z, 9, 50, 3)`, `fastNlMeansDenoising(z, None, 20, 7, 21)`. Ikki shovqin uchun PSNR/SSIM jadvali; xulosa.
23. **Qancha silliqlash?** Gauss shovqinli tasvirga `GaussianBlur`, $\sigma \in \{0,5; 0,75; 1; 1,5; 2; 3\}$: PSNR va SSIM; har birining optimal σ si. Nega ular farq qiladi?
24. **Tanlab ishlovchi median (amaliy: “oʻlik” piksellar).** `rng=default_rng(42)`, `m=rng.random(shape)`; `camera()` da `m<0.025` → 0, `m>0.975` → 255. `selective_median(img)` faqat 0/255 piksellarni `medianBlur`(, 3) bilan almashtiradi. Belgilangan piksellar soni, PSNR (shovqinli, toʻliq median, tanlab median), buzilmagan piksellarning oʻzgaray qolgan ulushi.
25. **Issiqlik tenglamasi.** `heat_diffuse(u, N, dt)`: $u \leftarrow u + \Delta t \text{filter2D}(u, L)$ (`BORDER_REFLECT`). $\Delta t = 0,2$, $N \in \{10, 25, 50, 100\}$: `GaussianBlur` ($\sigma = \sqrt{2N\Delta t}$) bilan oʻrtacha farq (chetdan 20 px ichkarida, absolyut va % da). Barqarorlik: $\Delta t = 0,25$ va $0,3$ da 200 qadam — $\max|u|$ va $|u| > 10^6$ boʻlgan birinchi qadam.
26. **Perona–Malik (sintetik).** `perona_malik(u, N=30, dt=0.2, k=20, c)` (4 yoʻnalish farqi, `np.pad(..., mode='edge')`). Sahna 128×128 : `x[:, 64:] = 200`, `x[32:96, 20:44] = 100`, `+ rng.normal(0, 15)` (seed 42). Gauss $\sigma = \sqrt{12}$ (`BORDER_REPLICATE`), P–M ($(c_1, k = 20)$, $(c_2, k = 20)$, $(c_1, k = 5)$, $(c_1, k = 80)$): PSNR, SSIM, chegara saqlanishi $100|\bar{u}_{:,64} - \bar{u}_{:,63}|/200$ %, qiymatlar oraligʻi.
27. **Perona–Malik (real, parametr tanlash).** `camera()+N(0, 15^2)` (seed 42, `clip`, `round`, `uint8`). $k \in \{10, 20, 30, 40\}$, $N \in \{10, 20, 40\}$, $\Delta t = 0,2$: PSNR/SSIM jadvali, eng yaxshi juftlik; eng yaxshi Gauss ($\sigma \in \{0,75; 1; 1,5\}$) va `bilateralFilter(9, 40, 3)` bilan solishtiring; `out/t27_pm_camera.png` (`[60:316, 150:406]`: shovqinli | Gauss | P–M).

- 28. Unsharp masking (defokusni tuzatish).** `blurred=GaussianBlur(camera(),(0,0),2); unsharp(img, sigma=2, lam) = f + λ(f - Gσ * f), λ ∈ {0,5;1;2;4}`: PSNR (`clip` va `round` dan keyin), `[0,255]` dan chiqqan piksellar soni, min/max. 14-topshiriqni 1D da tekshiring.
- 29. Wiener filtri (oracle).** `camera()+N(0,202) (float, seed 42). wiener_oracle(noisy, clean, sigma_n): SI = |FFT(I -  $\tilde{I}$ )|2, Sη = σn2 · HW, H = SI/(SI + Sη), H(0,0) = 1. PSNR ni eng yaxshi Gauss (σ ∈ {0,5;...;1,5}) bilan solishtiring;  $\tilde{H}$  va H > 0,5 ulushi. Nega bu “yuqori chegara”?`
- 30. CNN nuqtai nazari.** `F.conv2d (F.pad(...,mode='reflect'))` bilan σ = 1,5 Gauss (11 × 11) va S_x natijalarini `cv2.filter2D` bilan solishtiring. `torch.manual_seed(42); nn.Conv2d(3,64,3,stride=2,padding=1)` ning 224 × 224 dagi chiqish shakli va parametrlar soni.